

Smooth Snake Design Report

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Top-Level Design Description

Smooth Snake is a top-down arcade game where the player must guide the snake to eat apples. The goal is to eat as many apples as possible. When the snake eats an apple

it grows in length, making it more difficult to navigate around the screen as the game ends when if the snake accidentally eats itself.

Design Pillars

The below design pillars identify the types of fun which are key to the user experience

- Advancement & Completion
- Application of Ability
- Thrill of Danger

Advancement & Completion	Application of Ability	Thrill of Danger
Players will be satisfied as the score climbs and snake grows longer	The player has to maneuver around the snake's body to get apples. Players need to plan their path, avoiding trapping themselves.	The constant risk of hitting the snake's own body creates genuine tension, especially as the game progresses

Audience & Marketing

This is designed to be an offline game played via the Nintendo DS. It is a single player casual arcade game. Since this is a remake of a popular flash game from the early 2010's it will be aimed at a teen-adult audience who may remember the original.

The original game is no longer available online and this game is offered as an easy way to experience the core game mechanics that game offered once again.

Core Gameplay

The player maneuvers the snake around the game world by touching the snake's head and then dragging the stylus on the touchscreen in the desired direction.

An apple is randomly generated on screen. The player guides the snake's head towards the apple. Once the snake's head touches the apple, the apple disappears, and a crunch sound is played. The player's score is increased by 1. The snake grows a small but longer and another apple is randomly generated on screen.

The game gets more difficult as the snake eats more apples. As the snake will grow larger after each apple. The player may find they have to carefully navigate the snake out of a trap of its own making as, if the snake's head hits any other part of its body the snake eats itself and its game over. The game over screen displays your score and the best score.

Controls

Smooth Snake is designed for touchscreen input. The player touches the snake's head and drags in the desired direction to steer, giving an intuitive feel that suits the Nintendo DS touchscreen format.

Gameplay Balance & Pacing

The game's difficulty curve is built directly into its core mechanic. Each apple eaten increases the snake's length, which steadily reduces the amount of safe, open space the player has to maneuver in. This creates a natural escalation where early game movement is easy but late game movement demands precision since the player's own success becomes the source of increasing risk.

Touch screen controls were also tested to ensure the snake responded precisely so that skilled players could thread the snake through tight gaps created by its own body.

Tone & Aesthetics

The visual style of the game embraces a retro digital aesthetic, using a high-contrast green and black colour palette inspired by classic terminal displays and early arcade games. This minimal approach was a deliberate design choice suited to the Nintendo

DS hardware, keeping visuals clear on a small screen without relying on complex backgrounds.

Retro Digital



```
C:\Windows\system32\cmd.exe - color z
C:\Users\Prince>color z
Sets the default console foreground and background colors.

COLOR [attr]

attr          Specifies color attribute of console output

Color attributes are specified by TWO hex digits -- the first
corresponds to the background; the second the foreground. Each digit
can be any of the following values:

 0 = Black      8 = Gray
 1 = Blue       9 = Light Blue
 2 = Green      a = Light Green
 3 = Aqua       b = Light Aqua
 4 = Red        c = Light Red
 5 = Purple     d = Light Purple
 6 = Yellow     e = Light Yellow
 7 = White      f = Bright White

If no argument is given, this command restores the color to what it was
when CMD.EXE started. This value either comes from the current console
window, the /I command line switch or from the DefaultColor registry
value.
Press any key to continue . . .
```



Narrative

This is an offline singleplayer game without any explicit narrative. I do not intend to have dialog or characters speaking.

Business Model

Smooth Snake is a free-to-download Nintendo DS game with no monetisation model. It will be made freely available to all users with no ads and no premium tiers.

The game is being developed as a portfolio project to demonstrate skills in game design and development. Its primary value is in the player experience and the learning gained through building it, rather than revenue generation.